

Sales Data Transformation and Data Modeling

Objective:

This project focuses on transforming raw sales data into a structured and analysis-ready format using Power Query and building an efficient data model in Power BI. Data cleaning processes include handling missing values, removing duplicates, correcting data types, creating calculated columns, and optimizing data quality. A star schema data model is designed by establishing relationships between fact and dimension tables to improve performance and enable accurate reporting.

Dataset Description:

The dataset contains the following columns.

Column description:

Column Name	Description
Order Id	Order ID of the Product
Order Date	Order Date of the Product
Customer Name	Customer Name of the Product
State	State Name of the Customer
City	City Name of the Customer
Amount	Price of the Product
Profit	Profit of the Product
Quantity	Quantity of the Product
Category	Category of the Product
Sub-Category	Sub-Category of the Product
Month of Order Date	Order Date of the Sales
Target	Target for the Sales
Location	Newly Added City with State column
Profit Margin	Newly Calculated the Profit Margin
Profit Status	Newly Calculated the Profit status

Formulas Used:

1. To find the Profit Margin

Formula: Profit / Amount

2. To find the Profit Status

Formula:

- Profit < 0 → Loss
- Profit = 0 → Break-Even
- Profit > 0 → Profit

Conclusion:

The project demonstrates best practices in data preparation, transformation, and data modeling that support effective business intelligence and decision-making.

Newly merged table as Orders Data:

The screenshot shows a data tool interface with a table named "Orders Data". The table has the following columns: Order ID, Order Date, Customer Name, Location, Order Details Amount, Order Details Profit, Order Details Quantity, Order Details Category, and Order Details Sub-Category. The table contains 1,500 rows of data. The interface also shows a sidebar with icons for Structure, Data, and Relationships, and a top bar with various tool icons.

Order ID	Order Date	Customer Name	Location	Order Details Amount	Order Details Profit	Order Details Quantity	Order Details Category	Order Details Sub-Category
B-26099	30 March 2019	Bhishm	Mumbai, Maharashtra	₹ 207	37	4	Clothing	Hankerchief
B-26097	28 March 2019	Vini	Bangalore, Karnataka	₹ 14	5	1	Clothing	Hankerchief
B-26097	28 March 2019	Vini	Bangalore, Karnataka	₹ 19	8	2	Clothing	Hankerchief
B-26097	28 March 2019	Vini	Bangalore, Karnataka	₹ 97	12	2	Clothing	Hankerchief
B-26091	27 March 2019	Manju	Hyderabad, Andhra Pradesh	₹ 29	10	4	Clothing	Hankerchief
B-26093	27 March 2019	Sarita	Pune, Maharashtra	₹ 49	5	4	Clothing	Hankerchief
B-26086	26 March 2019	Mukesh	Chandigarh, Haryana	₹ 25	2	2	Clothing	Hankerchief
B-26086	26 March 2019	Bhavna	Gangtok, Sikkim	₹ 11	5	2	Clothing	Hankerchief
B-26089	26 March 2019	Kanak	Goa, Goa	₹ 27	4	1	Clothing	Hankerchief
B-26087	26 March 2019	Vandana	Simla, Himachal Pradesh	₹ 46	13	3	Clothing	Hankerchief
B-26077	22 March 2019	Divsha	Jaipur, Rajasthan	₹ 62	11	7	Clothing	Hankerchief
B-26079	22 March 2019	Hazel	Bangalore, Karnataka	₹ 18	3	2	Clothing	Hankerchief
B-26081	22 March 2019	Aarushi	Chennai, Tamil Nadu	₹ 24	11	3	Clothing	Hankerchief
B-26073	21 March 2019	Pournamasi	Indore, Madhya Pradesh	₹ 37	17	3	Clothing	Hankerchief
B-26073	21 March 2019	Pournamasi	Indore, Madhya Pradesh	₹ 122	11	4	Clothing	Hankerchief
B-26071	19 March 2019	Subhashree	Kashmir, Jammu and Kashmir	₹ 21	4	3	Clothing	Hankerchief
B-26070	18 March 2019	Shruti	Bangalore, Karnataka	₹ 14	7	2	Clothing	Hankerchief
B-26064	16 March 2019	Ankita	Mumbai, Maharashtra	₹ 61	3	4	Clothing	Hankerchief
B-26067	16 March 2019	Tulika	Bhopal, Madhya Pradesh	₹ 53	2	4	Clothing	Hankerchief
B-26058	11 March 2019	Aditya	Chandigarh, Punjab	₹ 212	97	7	Clothing	Hankerchief
B-26054	10 March 2019	Mayank	Mumbai, Maharashtra	₹ 246	98	5	Clothing	Hankerchief
B-26056	10 March 2019	Sonal	Patna, Bihar	₹ 206	18	4	Clothing	Hankerchief

Data Modelling:

